Tree Species Classification Using High-Resolution Satellite Imagery and Weakly Supervised Learning

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***Abstract*— Knowing vegetation type in an area is crucial for several applications, including ecology, land-use management, and infrastructure risk assessment. In combination with recent advancements in image processing, remote-sensing technology has been used to perform fast vegetation-type estimation and reduce the need for intensive and time-consuming field-based surveys. This article proposes a weakly supervised method based on deep learning to estimate tree species relying on multispectral high-resolution satellite images. We tested the approach against noisy labels, which often occur in real-world datasets. We validate our approach for a study area in Norway and Italy using images taken during different periods of the year. Our method significantly enhances the quality of the available forestry inventory dataset.**

***Index Terms*— Remote sensing, satellite imagery, tree species classification, weakly supervised learning.**

I. Introduction

HAVING a spatial tree species inventory is fundamental for a wide range of applications such as ecology and conservation [1], wildlife habitat mapping [2], sustainable land-use management [3], urban planning [4], as well as infrastructure monitoring and risk assessment [5], [6]. For example, power lines and roadway maintenance teams can make more effective tree trimming and cutting plans knowing the tree species [7], [8].

Conventional methods for documenting tree species are based on field surveys that are intensive, time-consuming, and often impractical in mountainous or inaccessible areas. However, remote-sensing technologies such as optical satellite images, light detection and ranging (LiDAR), and aerial images coming from helicopters or airborne vehicles have been explored over the last few decades to address such challenges.

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According to the review study conducted in [9], almost all the recent works combine a passive sensor (hyperspectral or multispectral optical sensor) with an active sensor, typically LiDAR. These kinds of sensors are often mounted on airborne vehicles. For example, in [10], the authors used a combination of airborne laser scanning (ALS) and color-infrared imagery to classify tree species in a mixed temperate forest in Poland. Airborne vehicles can provide detailed data over small areas, but they are still considered experimental [11] and, according to [12], currently, are not considered a stable platform to efficiently acquire systematic data to support forest inventories over large areas (>1000 ha). Airborne sensors are limited by flight time, large area coverage, and high operations costs. For example, LiDAR data is in the range of 62–240€ per km^2, while aerial imagery is approximately 35–62€ per km^2 [13].

Nowadays, several commercial satellite providers offer easy-accessible images for any part of the world with a high revisiting time (currently up to 0.25–0.5 m/pixel resolution). Furthermore, the drop in launching costs and the growing number of satellites and mini-satellites in orbit combined with high-quality sensors have reduced the cost of satellite imagery to less than 15€ per km^2 [13]. Satellite imagery is, therefore, currently the best tradeoff between acquisition price, quality, and revisiting frequency [14].

Remote-sensing-based tree species discrimination and classification evolved with methodological developments in the domain of statistical learning. In early studies, parametric approaches have been used (discriminant analysis, maximum likelihood) [15] while nonparametric approaches (decision trees, random forests, support vector machines, and neural networks) have gained more popularity in recent years [16], [17], [18]. Furthermore, the inclusion of texture information, mainly related to crown-internal shadows, foliage properties (size and branching) provided by the gray-level cooccurrence matrix (GLCM) [19] have been shown to result in improvement in the accuracy [20].

Recently, deep learning has become an effective tool for object detection and classification, especially in remote-sensing applications [21]. Among deep-learning techniques, convolutional neural networks (CNNs) have demonstrated high classification accuracy for digital images in the computer vision field [22]. Using cascades of learnable convolutional filters, it is possible to exploit the spatial relationship between pixels to provide information regarding the textures and shapes of trees [23]. In [24], the authors compared several

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neural network architectures using a hierarchical approach to detect dominant species in forests. In [25], the authors used WorldView-3 imagery and a Res-Unet deep-learning model for individual tree species classification. One of the most notable advantages of deep learning compared to other machine-learning methods is that deep learning does not require manual feature extraction.

In many current tree species classification works, labels are obtained via field investigation performed by research groups collecting samples using global positioning system (GPS) devices, which is a complex and time-intensive task. Other sources include existing tree reference datasets, which are often low resolution or noisy. However, machine-learning models are heavily affected by the quality of data labels. Unreliable or noisy labels severely degrade the generalization performance of such models as they can easily memorize corrupted labels and correspondingly degenerate their generalizations on unseen data [26]. Achieving a good generalization capability in the presence of noisy labels is still a key challenge [27].

This article proposes an automated approach to detect tree species using single high-resolution satellite images to lower costs and ease of implementation for practical reasons. The main contributions of this article are twofold.

1) From a methodological point of view, we propose a weakly supervised machine-learning approach to improve and refine forest inventory datasets. Our method relies on feature extraction from noisy labels and relabeling the data.
2) From a practical point of view, our approach refines the coarse and low-resolution tree species maps using one high-resolution satellite image. This process enhances the quality of the available forest inventories at a lower cost in scale in comparison to classic methods (e.g., sending ground-based crew or taking LiDAR or aerial images using helicopters and airplanes).

II. Study Area and Data Used

The main study area is a ≈25 km^2 region located in the Sogn og Fjordane county in Western Norway. We used a high-resolution satellite image from Maxar WorldView-2, acquired in the middle of May 2017. The image has a resolution of 0.5 m/pixel and has been ortho-rectified and pan-sharpened [28] by the provider.

The tree species reference dataset has been provided by the Norwegian Institute of Bioeconomy Research (NIBIO) [29], available and downloadable at [30]. The NIBIO dataset shows the forest distribution over the entire country. The dataset provides information such as the dominant tree species in forest stands. The term *forest stand* in forest inventory instructions defines a contiguous forested area sufficiently uniform in essential characteristics to distinguish it from adjacent communities.

The dominant tree species is defined on the basis of the estimated share of volume in a population as follows.

1) *Spruces*: If more than 50% of the trees are characterized as spruces, then the tree type's label is "spruces." The spruce group is mainly composed of Norway spruce (*Picea abies*), but also contains small portions of other spruce species such as Sitka spruce (*Picea sitchensis*).
2) *Pines*: If more than 50% of the trees are characterized as pines, then the tree type's label is "pines," specifically Scots pines (*Pinus sylvestris*).
3) *Deciduous*: If more than 50% of the trees are characterized as deciduous, then the tree type's label is "deciduous." The deciduous group is dominated by birch species (*Betula pubescens* and *Betula pendula*) but contains small portions of aspen (*Populus tremula*) and rowan (*Sorbus aucuparia*).

More detailed information on how the dataset has been created and validated can be found in [31].

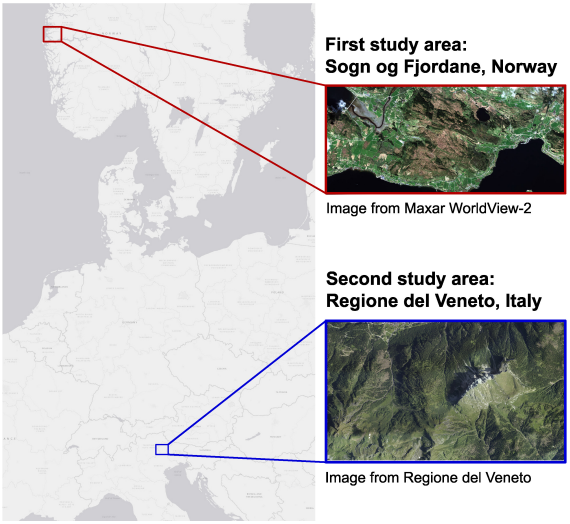


Fig. 1. Two study areas used. The first is located in the Sogn og Fjordane county, Western Norway. The second one is located in the Dolomites, in the Veneto region, Northern Italy.

A second study area has been selected in Italy, in the Dolomites, and part of the Veneto region. An image of ≈15 km^2 in the study area has been acquired in July 2019. The resolution is 0.5 m/px. The tree inventory dataset was acquired from *Regione del Veneto* at [32]. Similar to the NIBIO dataset, it divides the forest stands of the selected area into three groups: 1) *spruces*, mainly Norway spruces (*Picea abies*) which are also very common in the Dolomites; 2) larches (*Larix decidua*); and 3) deciduous, mainly gray alders (*Alnus incana*) and birches (*Betula pendula*).

The two study areas are shown in Fig. 1.

The tree inventory datasets are raster images with the same geographical extends as the satellite images where each pixel has a discrete value as: 0: no trees, 1: spruces, 2: pines/larches, 3: deciduous.

III. Methodology

We propose here a weakly supervised learning approach that produces a model capable of detecting the different species at almost tree level from a noisy low-resolution dataset.

To understand the reason and the advantages of such an approach, we first explain some issues present in both the

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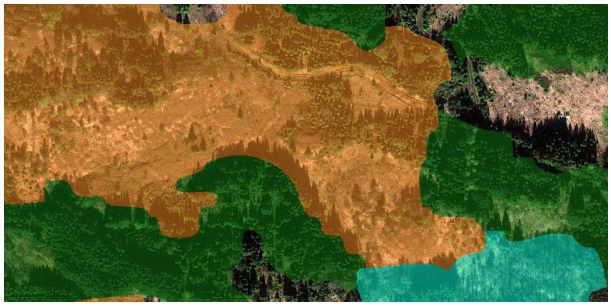


Fig. 2. Issues present in a typical tree species reference dataset. Some trees are not covered by the dataset and the boundaries between regions are often not precise.

datasets used for tree species and which may often be present in other real-world datasets. Fig. 2 shows some issues in the Norwegian tree species reference dataset.

We note, from Fig. 2, that some trees are not covered by the dataset even though they are visible from the satellite imagery. Moreover, the boundaries of the different forest stands are often chunky, tight, and not always precise. Finally, labels are assigned at a coarser level and hence some regions are marked even though they do not contain any trees. A conventional fully supervised segmentation procedure (i.e., extracting patches from both the satellite image and the tree dataset and directly training a model) will not handle these issues well and the model will not be able to learn properly. We propose a strategy to overcome these issues showing that we can learn from this dataset and refine it. Specifically, the method has the threefold advantage: detecting trees, assigning a label to those trees not initially covered by the dataset, and refining the boundaries between classes. The overall pipeline is sketched in Fig. 3 and each block is further described in the following.

A. Data Preprocessing

We first preprocess the multispectral satellite images. Pixel values in each band are scaled and stretched to the minimum and maximum values calculated, respectively, as the 2% and 98% percentile of the pixel distribution. This leads to higher contrast and better brightness uniformity among different images.

We use the normalized difference vegetation index (NDVI) as an additional channel in addition to the channels initially present. NDVI is a popular index in remote sensing for vegetation detection and it is defined as [33]

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{nir}} - \rho_{\mathrm{red}}}{\rho_{\mathrm{nir}} + \rho_{\mathrm{red}}} \tag{1}$$

where ρ_{red} and ρ_{nir} denote the spectral reflectance measurements acquired in the red (visible) and near-infrared regions, respectively. Green living plants are brighter in the near-infrared band due to chlorophyll's near-infrared high sensitivity. Each channel is then mapped into the same interval $(0, 1)$ to ease the training procedure.

B. Pixel-Wise Tree Segmentation Model

We design a tree segmentation algorithm. Given an input image, the corresponding output *treemap* is a single-channel pixel-wise mask. The segmentation model detects and labels the pixels that belong to (or are part of) a tree. We use an encoder–decoder-based architecture [34] as a segmentation model. The architecture is composed by a cascade of $[16, 32, 64, 128, 256]$ convolutional layers activated by a *relu* activation function, followed by a batch normalization layer and a Max Pooling layer. The architecture is shown in Fig. 4.

Binary cross entropy $\mathcal{L}$ is used as the loss function for training the network since only two labels are considered (0 for *no trees* and 1 for *trees*). To train the tree segmentation model, we use a manually created dataset for the considered study areas.

C. Tree Species Relabeling

The tree species reference we use in this study shows the dominant tree species in forest stands. However, it presents several issues (i.e., noisy or missing labels) as we showed in Fig. 2. The first step is to relabel the dataset to reduce the noise. This is because the species boundaries are coarse in the original dataset, and some trees are assigned to the same label as the surrounding trees, even if visually they are different species.

Therefore, we assume that trees from the same species have similar characteristics (features) in a feature space regardless of their assigned label. Furthermore, we assume that in most cases the given tree species inventory dataset $\mathbf{S}_{\mathrm{map}}$, provided by the Norwegian and Italian authorities, is correct. In other words, for example, the majority of trees marked as *spruces* are indeed *spruces*. The principle is sketched in Fig. 5. However, there are reservations regarding these assumptions which we discuss in Section V.

The implementation of the principle is divided into three steps.

1) Random Sample Extraction: We first multiply element-wise the tree reference dataset S_{map} (i.e., NIBIO dataset) with the *treemask* generated by the tree segmentation model in III-B. We denote the result $\mathbf{S}\mathrm{map}_{\mathrm{treemask}} \equiv \mathbf{S}\mathrm{map} \otimes \mathrm{treemask}$. In this way, fields, bare lands, roads, and any other nontree elements which are included in the original dataset are filtered out. Afterward, a defined number N of patches, each of size $s_p \times s_p$, are extracted randomly from the input satellite image, sampled among the three different families (i.e., *spruces*, *pines/larches*, and *deciduous*) of $\mathbf{S}\mathrm{map}_{\mathrm{treemask}}$. Each patch P_i, whose dimensions are $s_p \times s_p \times N_{ch}$, is naturally associated (yet not necessarily correct) with the label l taken from $\mathbf{S}\mathrm{map}_{\mathrm{treemask}}$.

$$P_{i=1,2,\ldots,N} \rightarrow l \in \{\text{spruces}, \text{pines/larches}, \text{deciduous}\}.$$

2) Feature Extraction: For each patch P_i, a feature vector $\mathcal{F}_i$ is extracted. Although features can be calculated in several ways, we use and compare two different techniques in this article.

The first approach is to hand-engineer the features. We create $\mathcal{F}$ as follows. The first components are the mean of each spectral band: $j = 1, 2, \ldots, N_{ch}$. In addition, texture signature operators [35] derived from the gray level cooccurrence matrix (GLCM) matrix are used. The GLCM can be thought of

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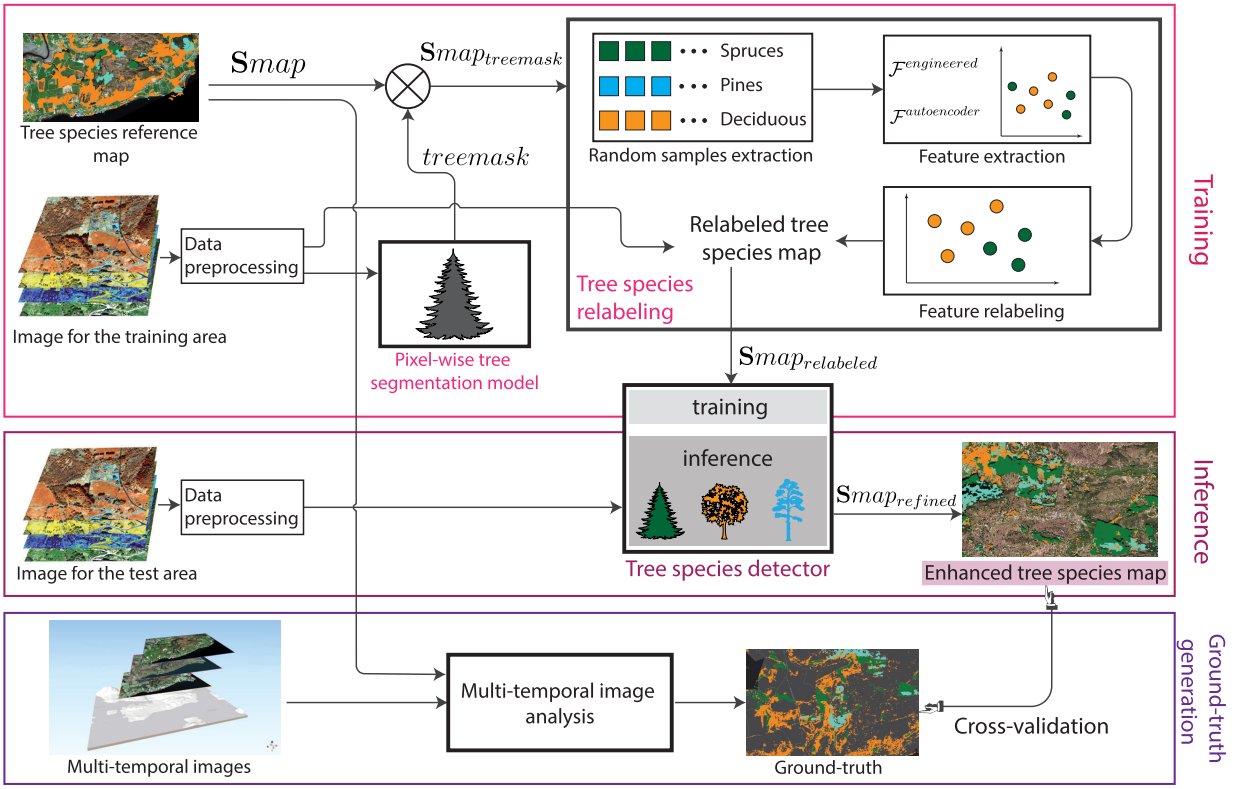


Fig. 3. Overall pipeline of the proposed approach. First, a binary tree segmentation model is trained to detect trees on the map. Random patches are extracted from the considered tree families (here spruces, pines, and deciduous), and for each patch, specific features are calculated. Such features are relabeled based on their position in the feature space. We then relabel the entire map, generating a prerefined version of the original tree species dataset. The output is used to train a deep-learning model. Once the model is trained, it can be used to generate refined versions of the original dataset at the tree level and with fuzzy boundaries.

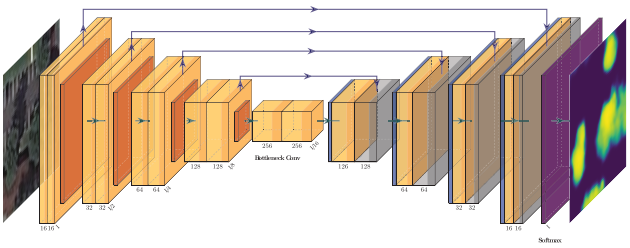


Fig. 4. Architecture used for trees segmentation model. Given a satellite image I as input, the corresponding output is a mask where trees are detected.

as a record of how often different combinations of pixel brightness values occur in imagery. While many operators exist, based on the principal component analysis performed in [18], we included the GLCM variance and mean over other operators, as they showed the two highest eigenvalues. With this approach, a feature vector is calculated from a generic patch P_i as follows:

$$P_i \rightarrow \mathcal{F}_i^{\text{engineered}} = \begin{bmatrix} \mathbb{E}_{j=1}^{N_{ch}} P_i(:,:,j) \\ \text{Variance(GLCM)} \\ \text{Mean(GLCM)} \end{bmatrix}. \quad (2)$$

The second approach is to take advantage of the deep-learning capabilities to extract features automatically using convolutional autoencoders [36]. In an autoencoder, an input image x is encoded through an encoder E_θ into a low-dimensional representation z (often denoted as the bottleneck layer or code of the autoencoder) through a cascade of convolutional kernels. The code can then be decoded by a decoder D_ϕ to reconstruct back the original image through upsampling operations. The autoencoder and its parameters θ and ϕ are trained to reconstruct as close as possible x in terms of the mean square error (MSE). Following this idea, we extract a feature vector $\mathcal{F}_i$ from the flattening of P_i's low-dimensional encoded representation:

$$P_i \rightarrow \mathcal{F}_i^{\text{autoencoder}} = E_\theta(P_i). \quad (3)$$

In this article, we used an encoder composed of a cascade of two convolutional layers activated by a *relu* activation function, followed by a batch normalization layer and a Max Pooling layer as shown in Fig. 6. The decoder architecture is symmetric.

3) Relabeling: Regardless of the method used for the feature extraction, a feature vector-label dataset $(\cap_{i=1}^{N}\{\mathcal{F}_i; l_i\})$ is generated, where $\mathcal{F} \in \mathbb{R}^d$. The value d is the dimension of the feature space, which depends on how many features are considered if the first approach is used (hand-engineered features) or on the autoencoder architecture if the second approach is used instead. Nevertheless, following the idea in Fig. 5, it is expected that trees that belong to different types are mapped into different regions of the $\mathbb{R}^d$ space. If it is

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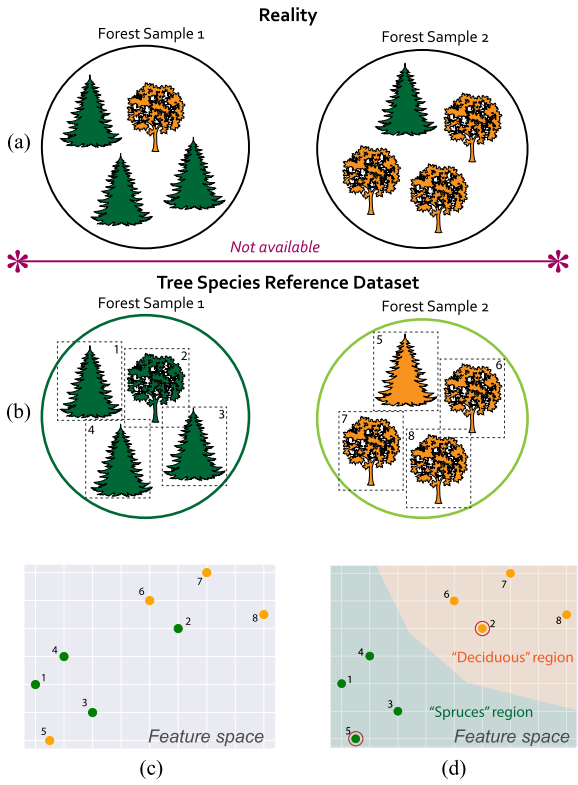


Fig. 5. Principle of the relabeling process. There are two forest stands in the example. (a) First has three spruces and one deciduous tree, while the second has three deciduous trees and one spruce. However, such information is not available. (b) In the available tree species reference dataset, all the trees in the first group are marked as *spruces* and all the trees in the second group are marked as *deciduous* regardless of their real type. (c) Trees of the same species should have similar characteristics (features) even if they are wrongly assigned different labels if mapped into a feature space. (d) Trees can be relabeled based on the position of their features in the feature space. The image shows a 2-D feature space with only spruce and deciduous classes for better visualization.

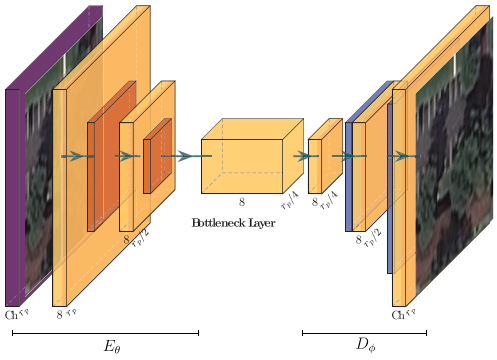


Fig. 6. Architecture used for the autoencoder. The feature vector $\mathcal{F}$ is extracted from the bottleneck layer.

possible to learn the mapping regions in the d-dimensional space, patches with the wrong label can be relabeled according to their position in the feature space. We refer to the process of assigning to each region of $\mathbb{R}^d$ a label as *semantic embedding*. The assumption that each tree type is distinguishable in the feature space holds, in theory. On the other hand, in practice, the cohesion and separation of each tree type's region in the feature space heavily depends on the "quality" of the feature extracted, which in turn depends on the quality of the input data as well as the power of the feature extraction method used.

We implement the semantic embedding using a majority voting scheme among the points. We first over-cluster the feature space with K clusters using the unsupervised clustering algorithm Kmeans. Over-clustering means to cluster space with more clusters than the number of groups the elements in the space belong to (which, in our case, is three, as the number of the considered tree types). Within each cluster $k = \{1, 2, \ldots, K\}$, we compute a score Φ defined as the number of points belonging to each tree type label l as follows:

$$\Phi_k^l = \text{Number of points in cluster } k \text{ with label } l. \quad (4)$$

We then assign to the cluster k the label with the highest score among Φ_k^l. Each point in the cluster k is assigned the label l set in the previous step. In this way, features "close" to each other in the feature space have the same label. Finally, patches are extracted sequentially from the whole area, and features are calculated based on one of the earlier described methods (engineered or autoencoder-based features). Each patch is relabeled according to the learned semantic embedding and the output is the relabeled map $\mathbf{S}\text{map}_{\text{relabeled}}$.

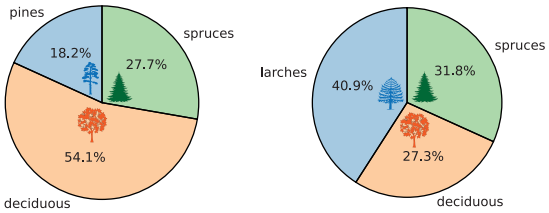
D. Tree Species Detector

Compared to the original tree type dataset **S**map, the relabeled map $\mathbf{S}\text{map}_{\text{relabeled}}$ has the following advantages: 1) each pixel belonging to a tree has a label; 2) nontrees are excluded from the dataset; and 3) trees visually similar have the same label. However, it has been generated patch-wise from the previous step, and therefore it is blocky. We design and train a CNN model to solve this issue. Such a procedure has two advantages. The first is to refine the tree species map $\mathbf{S}\text{map}_{\text{relabeled}}$ further, making the output smoother. The second one is to create a standalone model. Once it is trained, such a model can be used to directly estimate the tree species in the surrounding areas with the same tree species that has been trained without performing the relabeling process again. The architecture is similar to the one in Fig. 4 with the difference that the task is now a multilabel segmentation where the output is a three-label mask corresponding to spruces, pines/larches, and deciduous trees. The output is $\mathbf{S}\text{map}_{\text{refined}}$ in Fig. 3.

E. Ground-Truth Generation

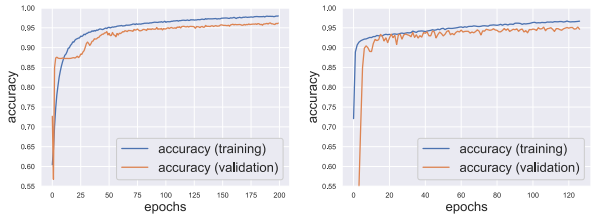
Finally, $\mathbf{S}\text{map}_{\text{refined}}$ is compared with the ground truth. Since the original tree inventory dataset **S**map cannot be used for comparison (because of the earlier discussed issues), we generate a ground truth by visual inspection, taking advantage of available acquired multitemporal images. For the Norwegian use case, we got two additional images: one from Airbus Pleiades-1 (four channels) acquired in September 2018 and another one from the Norwegian program *NorgeIBilder* [37]. The image is a three-channel image acquired in April 2019. For the Italian use case, we acquired two images from Planet, taken in April and October 2021, respectively. Similar to the process done in [18], interpretation was conducted using different imagery compositions of the available images at

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(a) (b)

Fig. 7. Distribution of pixels as percentages for (a) Norwegian study area and (b) Italian study area.



(a) (b)

Fig. 8. Training and validation accuracy for (a) Norway and (b) Italy.

different times, labeled by forest experts. The generation of such ground truth was a time-intensive, yet necessary task. Although a certain degree of uncertainty still exists, the obtained ground truth is more consistent than the original tree inventory. However, our approach is using the original raw reference dataset **S**map and, therefore, the ground-truth generation is not needed for the implementation.

IV. Results and Discussions

The proposed pipeline has been implemented in Python using the Keras/Tensorflow [38] libraries for the deep-learning part and the sklearn library [39] for the machine-learning part. Quantum Geographical Information System (QGIS) [40] has been used to integrate data and results. The testing platform is a computer equipped with an Intel i7 Core 10th Gen processor, 32 GB of RAM, and an Nvidia RTX 2080 Super as a graphics processing unit (GPU).

To validate the overall procedure, we divided each study area into two nonoverlapping regions A_1 and A_1 of similar size. Area A_1 is used for training and validation. This means that we calculate $\mathbf{S}\text{map}_{\text{treemask}}$ and $\mathbf{S}\text{map}_{\text{relabeled}}$ on A_1. Also, we train the deep-learning model only on area A_1. Afterward, we tested the trained model in A_2. The generated map $\mathbf{S}\text{map}_{\text{refined}}$ is compared with the ground truth. Ground truth has been created only for the testing region A_2. To avoid biases, A_1 and A_2 have been selected to have similar tree species distributions. Fig. 7 shows the distribution of pixels as percentages within each tree type group in the two study areas (e.g., Norway and Italy).

We notice that the datasets, as often happens in practice, are unbalanced. To overcome this issue, we sample the same number of patches per tree group in the *random sample extraction* step, when the patches are extracted to generate the feature space. Hence, a balanced dataset is used to create the feature space. Furthermore, when evaluating our final model in the testing area A_2, we use the F1-score as a metric to compare the different approaches. The F1-score is defined as the harmonic mean of the precision and recall and has been proved to be superior to accuracy when dealing with imbalanced classification problems [41].

A. Tree Segmentation Model

We first trained and tested the tree segmentation model used at the first stage of our methodology. The model is trained using a manually created binary mask of trees in the A_1 region. The dataset is split into a training and a validation set with an 80%–20% ratio, and the model described in III-B is trained. The Adam optimizer is used with a learning rate of 10^{-4}. An early stopping callback, which monitors the validation loss, keeps track of the loss, stopping the training to avoid overfitting. Fig. 8 shows the accuracy for the validation dataset, whose value stabilizes at ≈95%.

Once the model is trained, it can be used to generate the treemask. We multiply pixel-wise the original tree species dataset **S**map with treemask calculated from the trained tree segmentation model to get $\mathbf{S}\text{map}_{\text{treemask}} = \mathbf{S}\text{map} \otimes \text{treemask}$, as presented in Section III.

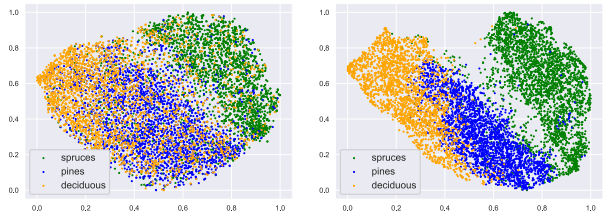
B. Relabeling Process

Once $\mathbf{S}\text{map}_{\text{treemask}}$ is created, randomly sampled patches of fixed size $s_p \times s_p$ are extracted from it. A small patch size makes it difficult to compute useful features out of it due to the lack of contextual information. On the other hand, a too large patch has the risk of incorporating different trees under the same label. We chose a patch size $s_p = 12 \times 12\ px$, which is equivalent to 6 × 6 m given the resolution of our satellite images. A balanced dataset of 2500 patches per tree type (spruces, pines, and deciduous) is extracted from the satellite image. We recall that each patch P has an associated label l derived from $\mathbf{S}\text{map}_{\text{treemask}}$.

Then, features are extracted using the two methods presented in Section III-C (i.e., engineered and autoencoder-based). The engineered features are calculated from each patch using 2. The GLCM matrix is calculated over each extracted patch P. We used the GLCM matrix computed with angles in the set $\{0; (\pi/2)\}$ and offset $\{1; 2\}$. Details on the implementation can be found in [42]. The autoencoder in Fig. 6 has been used to extract feature automatically using 3. For visualization purposes, we make use of the uniform manifold approximation and projection (UMAP) algorithm [43], to map the feature space $\mathbb{R}^d$ into $\mathbb{R}^2$. Fig. 9 shows the UMAP plots of the feature space generated via autoencoder for WorldView-2 [Fig 9(a)] with labels taken from $\mathbf{S}\text{map}_{\text{treemask}}$ and the multitemporal stacked image with labels taken from the ground truth [Fig. 9(b)].

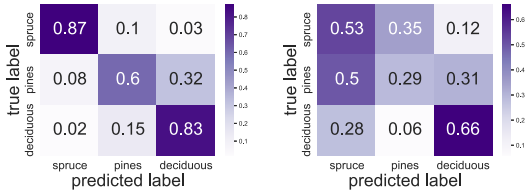
We recall that the feature space is a mathematical space where the points are the feature vectors $\mathcal{F} \in \mathbb{R}^d$ extracted from the patches $P_{i=1,\ldots,N}$. The UMAP algorithm map the relative position of the points in the higher-dimensional space into $\mathbb{R}^2$ positions while preserving the pairwise distance relations between points and the global structures. Therefore, the distance between points and the orientation of the

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(a) (b)

Fig. 9. UMAP plots of the feature space into $\mathbb{R}^2$. Feature space generated by the autoencoder from (a) WorldView-2 with labels taken from $\mathbf{S}\text{map}_{\text{treemask}}$ and (b) feature space generated by the autoencoder from the multitemporal stacked image (WorldView-Pleiades-NorgeIBilder) with labels taken from the ground truth.



(a) (b)

Fig. 10. Normalized confusion matrices (percentages) between predicted and true tree type for WorldView-2 imagery using (a) our learning procedure and autoencoder-based features and (b) traditional supervised approach without relabeling.

clusters do not have a geographical meaning. From the UMAP plots, we observe qualitatively that spruces, in particular, are more detached and separable from deciduous and pines in WorldView-2 imagery. From Fig. 9(b), we observe that the feature space generated using all the available images and the labels from the ground truth provided by III-E is smoother, with each tree type having its own region in the feature space. This supports our assumptions and the idea presented in Fig. 5. We then perform the relabeling process described in Section III. Points outside their semantic space are considered to have a wrong label and are then relabeled accordingly. We recall that the clustering is performed in the feature space $\mathbb{R}^d$. Once the semantic embedding has been performed in the feature space, patches of size $s_p \times s_p$ are extracted from the whole A_1 area, and the new learned labels are assigned, generating this way the output $\mathbf{S}\text{map}_{\text{relabeled}}$. It is worth mentioning that, besides the relabeling of the existing labels, this procedure also assigns to each tree that was not covered by the original tree reference dataset **S**map a new label.

C. Evaluation of the Inference Process

We train a CNN model using satellite images and the calculated $\mathbf{S}\text{map}_{\text{relabeled}}$ from both the engineered features and the autoencoder-based features as a training dataset. Similar to the tree segmentation model's training, we extract 15 000 tiles of 80×80 pixels (equal to 40×40 m given the resolution of our satellite images), randomly samples within A_1. We then train a model with the architecture and training parameters of the tree segmentation model. The output is three-channel, corresponding to the three tree types considered. Thus, a categorical cross-entropy loss function is used. To show the effectiveness of our approach, we test the performances using a CNN trained in a traditional supervised procedure. It consists in training the model using directly the original tree species dataset **S**map, without the relabeling process in our proposed pipeline. In addition, we train the model using $\mathbf{S}\text{map}_{\text{treemask}}$ and the highly refined ground truth.

1) Norwegian Study Area: First, we use confusion matrices to visualize, in particular, the classification matches per each category using the WorldView-2 image, as shown in Fig. 10.

Note how the confusion matrices with a standard approach are far from being diagonal [Fig. 10(b)]. Due to the intrinsic noise of the dataset, a model trained with a traditional procedure (i.e., without relabeling) is incapable of learning, and the performance easily degrades into low performance. In Fig. 10(a), we observe that while spruces and deciduous are correctly identified with high accuracy, pines are more problematic. There is a percentage of pines misclassified as deciduous trees.

TABLE I

F1-SCORES FOR EACH CLASS USING OUR PROPOSED APPROACH AND A TRADITIONAL TRAINING PROCEDURE

WorldView-2	**F1-score**		
	Spruces	*Pines*	*Deciduous*
Weakly-supervised + $\mathcal{F}^{\text{engineered}}$	0.799	0.506	0.675
Weakly-supervised + $\mathcal{F}^{\text{autoencoder}}$	**0.883**	**0.648**	**0.761**
Supervised from $\mathbf{S}map$	0.458	0.322	0.531
Supervised from $\mathbf{S}map_{treemask}$	0.492	0.441	0.586

Table I provides the F1-score per each class for WorldView-2 using different approaches.

Learning from $\mathbf{S}\text{map}_{\text{treemask}}$ slightly improves the performances compared to learning directly from **S**map. This is due to the fact that in $\mathbf{S}\text{map}_{\text{treemask}}$, the nontree elements have been filtered out. However, the coarse boundaries between forest stands are still the same as **S**map and many pixels remain unlabeled. We note how using the relabeling step, before training the model, greatly boosts the performance with an average enhancement of the F1-score of 78% for the WorldView-2 imagery compared to not using our relabeling step. Finally, not surprisingly learning directly from the ground truth leads to the best results (F1-score: 0.915, 0.797, 0.864). However, this is not applicable in practice as the ground truth, which has been created by visual inspection of multitemporal high-resolution images, is not available in general. It is worth mentioning that engineered features are challenging to design. For example, texture operators can be applied to each channel separately or any combination of them. Manually engineering this considerable amount of combinations is impractical. Features extracted from autoencoders are, by definition, extracted automatically from the network instead.

Finally, in Table II, we show the computational running times of the different methods.

The high running time of the features extraction method lies in the central processing unit (CPU) implementation of the sklearn library. Features are extracted within a for-loop (which is usually slow in Python), called several times in the main script. On the other hand, autoencoders are running on

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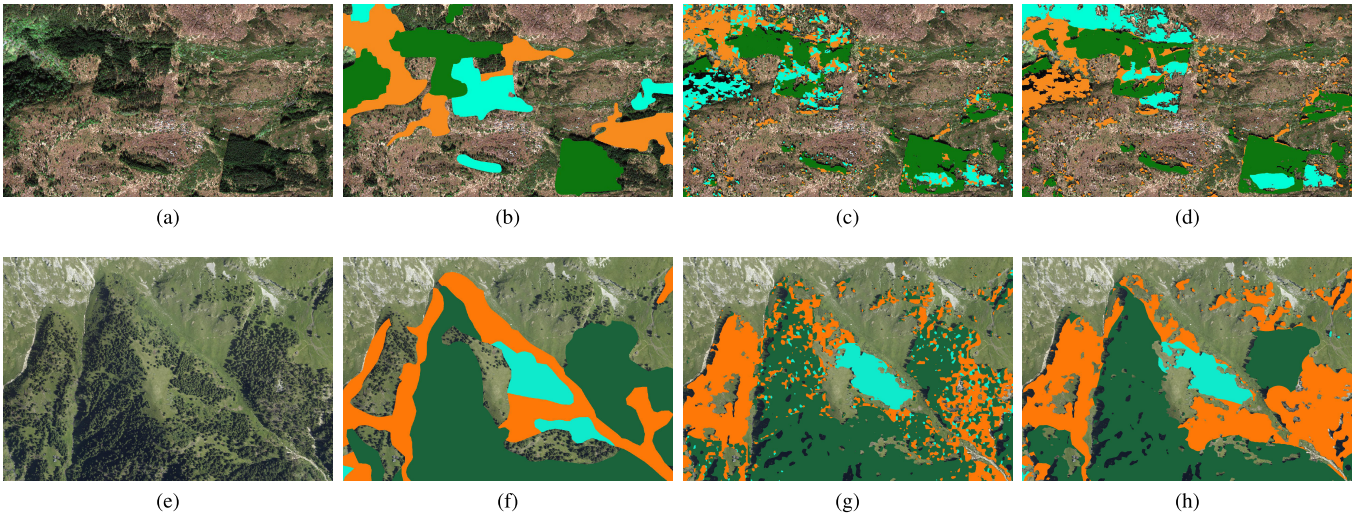


Fig. 11. Sample of the WorldView imagery in Norway (top row) and Italy (bottom row). (a) and (e) Optical image. (b) and (f) Starting tree species dataset **S**map. (c) and (g) Refined tree species map $\mathbf{S}\text{map}_{\text{refined}}$. (d) and (h) Ground truth. Note the issues present in the original dataset. Despite these, the output generated via our approach can automatically capture new trees and refine better the forest stands. Legend: spruces (green), pines/larches (cyan), and deciduous (orange).

TABLE II
COMPUTATIONAL RUNNING TIMES FOR THE DIFFERENT APPROACHES

Routine	Running time	
	Training	Inference
Engineered features extraction	—	491 sec
Autoencoder-based feature extraction	28.4 sec	5.3 sec
Tree species detector	36 min	8.2 sec

TABLE III
F1-SCORES FOR EACH CLASS USING OUR PROPOSED APPROACH AND A TRADITIONAL TRAINING PROCEDURE FOR THE ITALIAN USE CASE

Imagery and Feature Type	F1-score		
	Spruces	*Larches*	*Deciduous*
Weakly-supervised + $\mathcal{F}^{\text{engineered}}$	0.381	0.387	0.442
Weakly-supervised + $\mathcal{F}^{\text{autoencoder}}$	**0.422**	**0.450**	**0.494**
Supervised from $\mathbf{S}map$	0.261	0.266	0.357
Supervised from $\mathbf{S}map_{treemask}$	0.312	0.304	0.408

GPU using the TensorFlow implementation, making much fast the process of feature extraction. Furthermore, autoencoders' architecture contains much less parameters (2323 learnable weights) compared to the segmentation model (1 946 993 learnable weights), making the training procedure much faster.

2) Italian Study Area: For further validation, we implement the approach in another study area located in Italy. Table III provides the resulting F1-scores per each class.

We note that our relabeling approach improves the results compared to the traditional approach, confirming the outcomes found in the first use case. Learning directly from the ground truth leads again to the best results (F1-score: 0.531, 0.495, and 0.585). However, we also notice that the F1-scores, in this case, are generally lower than in the previous case study. This is due to the fact that the image has been acquired in summer, which is the season that minimizes the seasonality changes between tree species. We provide further discussion in V.

In both cases, our methodology can properly train a model through a two-step relabeling procedure, showing clear benefits compared to traditional approaches without the relabeling step. It can generate a more detailed version of the original tree species dataset. A visual example is shown in Fig. 11.

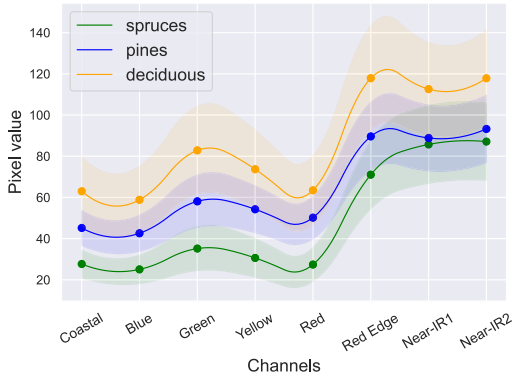
We note how the generated output is much more refined and semantically more coherent than the original dataset. Our procedure can greatly enhance the quality of available datasets for tree species.

V. PRACTICAL CONSIDERATIONS AND LIMITATIONS

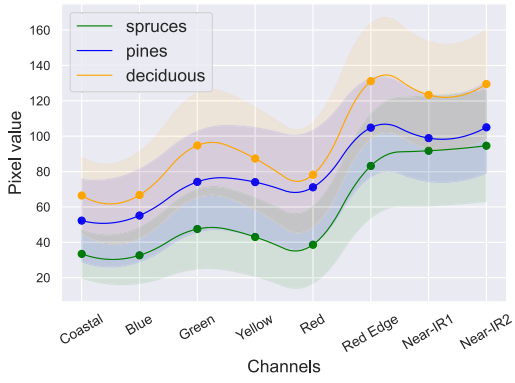
Correctly recognizing tree species from satellite images is a difficult task, especially when available tree inventory is noisy and not precise enough, like in this study. However, there is another consideration. Available imagery might not be sufficient to recognize the different species effectively. To demonstrate this problem, we extracted random patches from the three species, labeled according to the original tree inventory dataset **S**map and from the ground truth, respectively, in the Norwegian study area. We compute and plot the mean (as a solid line) and standard deviation (as a confidence interval line plot) per channel in Fig. 12.

We observe that in the Norwegian use case (WorldView imagery from May), if we sample from the ground truth [Fig. 12(a)], each tree species has a specific spectral curve. On the other hand, the spectral curves taken from **S**map [Fig. 12(b)] are more mixed. This shows that the ground truth is more consistent than the starting tree species inventory **S**map. However, we recall that the ground truth is not available in practice in most cases. It is worth mentioning that spectral curves only show pixel intensity in the different spectra but other contextual information (related, for instance, to texture) that can potentially be used to discriminate between species is missing. This suggests and confirms several studies [44] that the number of channels available (i.e., the amount of information we can extract from satellite images) and the seasonality play an important role indeed.

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(a)



(b)

Fig. 12. Spectral curves per channel with (a) patches sampled from the ground truth and (b) patches sampled from the **S**map (NIBIO dataset).

Spruces have been found to be easily identified in Norway. However, compared to spruces, which have large, dark green canopies, pines have more scattered and pruned canopies with little foliage, making them more difficult to distinguish from deciduous trees. The fact that pines are the most problematic type is also confirmed by Fig. 9 where we observe that their features generally lay between the spruces and deciduous' space, making it less separable. In Italy, on the other hand, we found spruces and larches to be less distinguishable. This is because they have nearly the same canopy structure. However, larches are one of the few conifers that lose their needles in the autumn. Therefore, it is desirable to align the time of image acquisition with the phenological cycle of the species under investigation to increase the performance of tree species classification.

As presented in Section III, we use two main assumptions to facilitate our framework development. The first assumption is that trees of the same species share the same position in the feature space. However, there is a possibility that two or more different tree species have similar visual characteristics. Therefore, we suggest, in regions where such tree species exist, to combine classic ground-based approaches in addition to satellite surveys. The second assumption is that the majority of trees have the right label in the given tree species inventory dataset **S**map. However, generating tree inventories at the national or regional level is a time-consuming and challenging task that has some uncertainties. In regions where existing tree inventories have low accuracy, it is possible to manually label part of the satellite images by experts on a limited scale.

While the proposed methodology works with any number of images, we were particularly interested in evaluating the method when only one multispectral satellite image is used. Acquiring multitemporal images has some advantages in terms of performance (because more information can be extracted), but also exhibits some potential drawbacks. Multitemporal high-resolution images can be expensive. In addition, land use can change and large portions of trees can be cut down between two frames, making the analysis more complicated to automate.

Furthermore, hyperspectral imaging is an option to discriminate even further different types of trees. Hyperspectral images contain multiple (typically between 64 and 256) continuous narrow bands, providing significant levels of detail, which allow for the distinction of fine spectral variations among tree species [45]. However, the main limitations of such images are the costs, complexity, and their availability in different locations. Recently, some hyperspectral sensors have been launched, including earth observing-1 (EO-1) from National Aeronautics and Space Administration (NASA) and Project for On-Board Autonomy-1 (PROBA-1) from European Space Agency (ESA), spanning from 60 to 200 spectral bands with a 30-m resolution. To our knowledge, no satellite providers offer high-resolution hyperspectral images (i.e., 1-m or less). The current studies on hyperspectral images are limited to airborne scanning, which faces the challenges introduced in Section I.

Different available approaches exist, each with specific advantages and disadvantages (quality, data availability, coverage, storage constraints, costs, etc.). We find that our method can serve as a complementary source to the other approaches, providing a high potential for added satellite-based observability with minimal costs and high availability.

VI. Conclusion

This article has proposed a framework to identify tree species using high-resolution satellite images and a noisy reference forestry survey dataset. The proposed methodology relies on feature extraction from samples in the tree reference dataset to learn the characteristic properties of each class. The features are relabeled based on their position in the feature space, and a prerefined map is generated using an additional tree segmentation model. A deep-learning model is then trained with the prerefined map to further smooth the prerefined map. The strength of our approach is that refined versions of tree species inventories, often available at regional scales, can be generated automatically from single images as input, making the approach practical and relatively cheap. Multitemporal images and different case studies have been used for cross-validation. Our evaluation shows that the proposed methodology exhibits clear benefits compared to traditional training procedures. Our method can be helpful as a complementary source, providing a high potential for added observability with minimal costs and high availability for everyone. Future work will be toward improving further the performances, especially with trees with less foliage.

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In addition, we plan to improve the methodology by detecting outliers (trees that do not belong to any of the specified species) in the feature space and testing the approach with more tree species.

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